Human Physiology

The reductionist approach to the study of life forms resulted in increasing use of physico-chemical concepts and techniques, most studies employing either a surviving tissue model or straightaway cell-free systems.

Where that led

- an explosion of knowledge resulted in molecular biology
- molecular physiology became almost synonymous with biochemistry and biophysics
- it is now increasingly realised that neither a purely organismic approach nor a purely reductionistic molecular approach would reveal the truth about biological processes

Systems biology

- it makes us believe that all living phenomena are emergent properties, due to interaction among components of the system under study
- regulatory networks of molecules, supra molecular assemblies, cells, tissues, organisms and indeed populations and communities each create emergent properties

What this unit covers

- major human physiological processes such as exchange of gases, blood circulation, locomotion and movement, described in cellular and molecular terms
- the last two chapters point to the coordination and regulation of body events at the organismic level

In this unit

Ch 14 Breathing and Exchange of Gases

Ch 15 Body Fluids and Circulation

Ch 16 Excretory Products and their Elimination

Ch 17 Locomotion and Movement

Ch 18 Neural Control and Coordination

Ch 19 Chemical Coordination and Integration

Alfonso Corti, 1822 to 1888

- an Italian anatomist, born in 1822
- he began his scientific career studying the cardiovascular systems of reptiles
- later he turned his attention to the mammalian auditory system
- he died in the year 1888

The organ of Corti

- in 1851 he published a paper describing a structure located on the basilar membrane of the cochlea
- it contains hair cells that convert sound vibrations into nerve impulses
- the structure is named the organ of Corti after him

Why we breathe

- oxygen is utilised by organisms to indirectly break down simple molecules such as glucose, amino acids and fatty acids, to derive energy to perform various activities
- carbon dioxide, which is harmful, is also released during these catabolic reactions
- so O_2 has to be continuously provided to the cells and the CO_2 produced by them has to be released out
- this process of exchange of O_2 from the atmosphere with CO_2 produced by the cells is called breathing, commonly known as respiration

14.1 Respiratory organs

Among the invertebrates

- mechanisms of breathing vary among different groups of animals, depending mainly on their habitats and levels of organisation
- lower invertebrates such as sponges, coelenterates and flatworms exchange O_2 with CO_2 by simple diffusion over their entire body surface
- earthworms use their moist cuticle, and insects have a network of tubes, the tracheal tubes, to transport atmospheric air within the body
- special vascularised structures called gills, that is branchial respiration, are used by most aquatic arthropods and molluscs

Among the vertebrates

- vascularised bags called lungs, that is pulmonary respiration, are used by terrestrial forms for the exchange of gases
- fishes use gills, whereas amphibians, reptiles, birds and mammals respire through lungs
- amphibians such as frogs can respire through their moist skin also, which is cutaneous respiration

External nostrils leading through the nasal chamber to the pharynx

Larynx with its epiglottis, then the trachea

Bronchus dividing into bronchioles

Alveoli at the ends of the terminal bronchioles

Lungs within the pleural membranes, with pleural fluid between

Diaphragm below, and the cut end of a rib alongside

Figure 14.A Diagrammatic view of human respiratory system (sectional view of the left lung is also shown)

14.2 The human respiratory system

From nostril to trachea

- we have a pair of external nostrils opening out above the upper lips, leading to a nasal chamber through the nasal passage
- the nasal chamber opens into the pharynx, a portion of which is the common passage for food and air, and the pharynx opens through the larynx region into the trachea
- the larynx is a cartilaginous box which helps in sound production, hence is called the sound box, and during swallowing the glottis can be covered by a thin elastic cartilaginous flap, the epiglottis, to prevent the entry of food

From trachea to alveoli

- the trachea is a straight tube extending up to the mid-thoracic cavity, which divides at the level of the 5th thoracic vertebra into a right and left primary bronchi
- each bronchus undergoes repeated divisions to form the secondary and tertiary bronchi and bronchioles, ending in very thin terminal bronchioles
- the trachea, the primary, secondary and tertiary bronchi, and the initial bronchioles are supported by incomplete cartilaginous rings
- each terminal bronchiole gives rise to a number of very thin, irregular-walled and vascularised bag-like structures called alveoli

The lungs and the pleura

- the branching network of bronchi, bronchioles and alveoli comprise the lungs
- we have two lungs, covered by a double layered pleura with pleural fluid between them, which reduces friction on the lung surface
- the outer pleural membrane is in close contact with the thoracic lining

14.3 Conducting and exchange parts

Two parts of the system

- the inner pleural membrane is in contact with the lung surface
- the part starting with the external nostrils up to the terminal bronchioles constitutes the conducting part, whereas the alveoli and their ducts form the respiratory or exchange part
- the conducting part transports the atmospheric air to the alveoli, clears it of foreign particles, humidifies it and brings it to body temperature
- the exchange part is the site of actual diffusion of O2 and CO2 between blood and atmospheric air

The thoracic chamber

- the lungs are situated in the thoracic chamber, which is anatomically an air-tight chamber
- it is formed dorsally by the vertebral column, ventrally by the sternum, laterally by the ribs, and on the lower side by the dome-shaped diaphragm
- the anatomical setup is such that any change in the volume of the thoracic cavity will be reflected in the pulmonary cavity, an arrangement essential for breathing since we cannot directly alter the pulmonary volume

14.4 The steps of respiration

Five steps

- breathing, or pulmonary ventilation, by which atmospheric air is drawn in and CO2 rich alveolar air is released out
- diffusion of gases across the alveolar membrane
- transport of gases by the blood, and diffusion of O2 and CO2 between blood and tissues
- utilisation of O2 by the cells for catabolic reactions and the resultant release of CO2, that is cellular respiration

14.5 Mechanism of breathing

Pressure gradients

- breathing involves two stages, inspiration during which atmospheric air is drawn in, and expiration by which the alveolar air is released out
- the movement of air into and out of the lungs is carried out by creating a pressure gradient between the lungs and the atmosphere
- inspiration can occur if the intra-pulmonary pressure is less than the atmospheric pressure, and expiration takes place when it is higher
- the diaphragm and a specialised set of muscles, the external and internal intercostals between the ribs, help in generating such gradients

Inspiration, with the diaphragm contracting and the ribs lifting

Thoracic volume increasing, pulmonary pressure falling, air drawn in

Expiration, with diaphragm and intercostals relaxing

Thoracic volume falling, pulmonary pressure rising, air expelled

Figure 14.B Mechanism of breathing showing : (a) inspiration (b) expiration

Inspiration

- it is initiated by the contraction of the diaphragm, which increases the volume of the thoracic chamber in the antero-posterior axis
- the contraction of the external inter-costal muscles lifts up the ribs and the sternum, increasing the volume of the thoracic chamber in the dorso-ventral axis
- the overall increase in thoracic volume causes a similar increase in pulmonary volume, which decreases the intra-pulmonary pressure to less than atmospheric, forcing air from outside into the lungs

Expiration

- relaxation of the diaphragm and the inter-costal muscles returns the diaphragm and sternum to their normal positions, reducing the thoracic volume and thereby the pulmonary volume
- this leads to an increase in intra-pulmonary pressure to slightly above atmospheric, causing the expulsion of air from the lungs
- we have the ability to increase the strength of inspiration and expiration with the help of additional muscles in the abdomen
- on an average a healthy human breathes 12 to 16 times per minute, and the volume of air involved can be estimated using a spirometer, which helps in clinical assessment of pulmonary functions

14.6 Respiratory volumes

Three volumes

- tidal volume is the volume of air inspired or expired during a normal respiration, approximately 500 mL, so a healthy man can inspire or expire approximately 6000 to 8000 mL of air per minute
- inspiratory reserve volume is the additional volume a person can inspire by a forcible inspiration, averaging 2500 mL to 3000 mL
- expiratory reserve volume is the additional volume a person can expire by a forcible expiration, averaging 1000 mL to 1100 mL

Residual volume

- the volume of air remaining in the lungs even after a forcible expiration
- this averages 1100 mL to 1200 mL
- by adding up a few respiratory volumes one can derive various pulmonary capacities, which can be used in clinical diagnosis

14.7 Pulmonary capacities

Inspiratory, expiratory, functional residual

- inspiratory capacity is the total volume of air a person can inspire after a normal expiration, including tidal volume and inspiratory reserve volume
- expiratory capacity is the total volume a person can expire after a normal inspiration, including tidal volume and expiratory reserve volume
- functional residual capacity is the volume of air that will remain in the lungs after a normal expiration, including expiratory reserve volume and residual volume

Vital and total lung capacity

- vital capacity is the maximum volume of air a person can breathe in after a forced expiration, including expiratory reserve volume, tidal volume and inspiratory reserve volume, or the maximum volume a person can breathe out after a forced inspiration
- total lung capacity is the total volume of air accommodated in the lungs at the end of a forced inspiration, including residual volume, expiratory reserve volume, tidal volume and inspiratory reserve volume, that is vital capacity plus residual volume

14.8 Exchange of gases

Where and how

- alveoli are the primary sites of exchange of gases, and exchange also occurs between blood and tissues
- O2 and CO2 are exchanged at these sites by simple diffusion, mainly based on pressure or concentration gradient
- solubility of the gases, as well as the thickness of the membranes involved in diffusion, are also important factors that can affect the rate of diffusion
- the pressure contributed by an individual gas in a mixture of gases is called partial pressure, represented as pO2 for oxygen and pCO2 for carbon dioxide

Table 14.A Partial Pressures (in mm Hg) of Oxygen and Carbon dioxide at Different Parts Involved in Diffusion in Comparison to those in Atmosphere

Respiratory Gas	Atmospheric Air	Alveoli	Blood (Deoxygenated)	Blood (Oxygenated)	Tissues
O2	159	104	40	95	40
CO2	0.3	40	45	40	45

Alveolus with its thin squamous epithelium

Pulmonary capillary alongside it

Oxygen passing in, carbon dioxide passing out

Figure 14.C Diagrammatic representation of exchange of gases at the alveolus and the body tissues with blood and transport of oxygen and carbon dioxide

The alveolar epithelium

The endothelium of the alveolar capillary

The basement substance lying between them

Figure 14.D A Diagram of a section of an alveolus with a pulmonary capillary.

The gradients

- the table shows a concentration gradient for oxygen from alveoli to blood and from blood to tissues
- a gradient is present for CO_2 in the opposite direction, that is from tissues to blood and from blood to alveoli
- as the solubility of CO_2 is 20 to 25 times higher than that of O_2, the amount of CO_2 that can diffuse through the diffusion membrane per unit difference in partial pressure is much higher than that of O_2

14.9 The diffusion membrane

Three layers

- the thin squamous epithelium of alveoli
- the endothelium of alveolar capillaries
- the basement substance between them, composed of a thin basement membrane supporting the squamous epithelium and the basement membrane surrounding the single layer endothelial cells of capillaries
- its total thickness is much less than a millimetre, so all the factors in our body are favourable for diffusion of O_2 from alveoli to tissues and of CO_2 from tissues to alveoli

Percentage saturation of haemoglobin plotted against partial pressure of oxygen

A sigmoid curve rising steeply then levelling off

High pO2, low pCO2 and lower temperature favour oxyhaemoglobin

Low pO2, high pCO2 and higher temperature favour dissociation

Figure 14.E Oxygen dissociation curve

14.10 Transport of gases

How much travels how

- blood is the medium of transport for O2 and CO2
- about 97 per cent of O2 is transported by RBCs in the blood, the remaining 3 per cent carried in a dissolved state through the plasma
- nearly 20 to 25 per cent of CO2 is transported by RBCs, whereas 70 per cent is carried as bicarbonate, and about 7 per cent in a dissolved state through plasma

14.11 Transport of oxygen

Haemoglobin

- haemoglobin is a red coloured iron containing pigment present in the RBCs, and O2 can bind with it in a reversible manner to form oxyhaemoglobin
- each haemoglobin molecule can carry a maximum of four molecules of O2
- binding is primarily related to the partial pressure of O2, while partial pressure of CO2, hydrogen ion concentration and temperature are the other factors that can interfere
- a sigmoid curve is obtained when percentage saturation of haemoglobin with O2 is plotted against pO2, called the oxygen dissociation curve

Where it binds and where it lets go

- in the alveoli, where there is high pO2, low pCO2, lesser hydrogen ion concentration and lower temperature, the factors are all favourable for the formation of oxyhaemoglobin
- in the tissues, where low pO2, high pCO2, high hydrogen ion concentration and higher temperature exist, the conditions favour dissociation of oxygen from oxyhaemoglobin
- so O2 gets bound to haemoglobin at the lung surface and gets dissociated at the tissues
- every 100 mL of oxygenated blood can deliver around 5 mL of O2 to the tissues under normal physiological conditions

14.12 Transport of carbon dioxide

As carbamino-haemoglobin

- CO2 is carried by haemoglobin as carbamino-haemoglobin, about 20 to 25 per cent of it
- this binding is related to the partial pressure of CO2, and pO2 is a major factor which could affect it
- when pCO2 is high and pO2 is low, as in the tissues, more binding of carbon dioxide occurs

As bicarbonate

- when pCO_2 is low and pO_2 is high, as in the alveoli, dissociation of CO_2 from carbamino-haemoglobin takes place, so CO_2 bound to haemoglobin from the tissues is delivered at the alveoli
- RBCs contain a very high concentration of the enzyme carbonic anhydrase, and minute quantities are present in the plasma too, the enzyme facilitating the reaction $CO_2 + H_2O \rightarrow H_2CO_3 \rightarrow HCO_3^- + H^+$ in both directions
- at the tissue site, where partial pressure of CO_2 is high due to catabolism, CO_2 diffuses into blood and forms bicarbonate and hydrogen ions
- at the alveolar site, where pCO_2 is low, the reaction proceeds in the opposite direction, so CO_2 trapped as bicarbonate at the tissue level is released out, every 100 mL of deoxygenated blood delivering approximately 4 mL of CO_2 to the alveoli

14.13 Regulation of respiration

The centres

- human beings have a significant ability to maintain and moderate the respiratory rhythm to suit the demands of the body tissues, and this is done by the neural system
- a specialised centre in the medulla region of the brain, the respiratory rhythm centre, is primarily responsible for this regulation
- another centre in the pons region, the pneumotaxic centre, can moderate its functions, its neural signal reducing the duration of inspiration and thereby altering the respiratory rate

The chemosensitive area

- a chemosensitive area is situated adjacent to the rhythm centre, highly sensitive to CO_2 and hydrogen ions
- an increase in these substances activates this centre, which signals the rhythm centre to make the necessary adjustments by which they can be eliminated
- receptors associated with the aortic arch and carotid artery also recognise changes in CO_2 and hydrogen ion concentration and send signals to the rhythm centre
- the role of oxygen in the regulation of respiratory rhythm is quite insignificant

14.14 Disorders of the respiratory system

Asthma and emphysema

- asthma is a difficulty in breathing causing wheezing, due to inflammation of bronchi and bronchioles
- emphysema is a chronic disorder in which alveolar walls are damaged, decreasing the respiratory surface, one of the major causes being cigarette smoking

Occupational respiratory disorders

- in certain industries, especially those involving grinding or stone-breaking, so much dust is produced that the defense mechanism of the body cannot fully cope with the situation
- long exposure can give rise to inflammation leading to fibrosis, that is proliferation of fibrous tissues, and thus serious lung damage
- workers in such industries should wear protective masks

14.15 Summary

The system and the steps

- cells utilise oxygen for metabolism and produce energy along with substances such as carbon dioxide, which is harmful, and animals have evolved different mechanisms for the transport of oxygen to the cells and removal of carbon dioxide
- we have a well developed respiratory system comprising two lungs and associated air passages
- the first step is breathing, by which atmospheric air is taken in and alveolar air released out, followed by exchange of gases between deoxygenated blood and alveoli, transport by blood, exchange between oxygenated blood and tissues, and utilisation of O_2 by the cells
- inspiration and expiration are carried out by creating pressure gradients with the help of specialised muscles, the intercostals and diaphragm, and the volumes involved can be estimated with a spirometer and are of clinical significance

Diffusion and transport

- exchange at the alveoli and tissues occurs by diffusion, the rate depending on the partial pressure gradients of O_2 and CO_2, their solubility, and the thickness of the diffusion surface
- these factors favour diffusion of O_2 from the alveoli to the deoxygenated blood and from the oxygenated blood to the tissues, and of CO_2 in the opposite direction
- oxygen is transported mainly as oxyhaemoglobin, binding where pO_2 is higher and dissociating at the tissues where pO_2 is low and pCO_2 and hydrogen ion concentration are high
- nearly 70 per cent of carbon dioxide is transported as bicarbonate with the help of carbonic anhydrase, and 20 to 25 per cent by haemoglobin as carbamino-haemoglobin

Regulation

- respiratory rhythm is maintained by the respiratory centre in the medulla region of the brain
- a pneumotaxic centre in the pons region and a chemosensitive area in the medulla can alter the respiratory mechanism

14.16 Exercises

From this chapter

1. Define vital capacity. What is its significance?
2. State the volume of air remaining in the lungs after a normal breathing.
3. Diffusion of gases occurs in the alveolar region only and not in the other parts of the respiratory system. Why?
4. What are the major transport mechanisms for carbon dioxide? Explain.
5. What will the partial pressures of oxygen and carbon dioxide be in the atmospheric air compared to those in the alveolar air? (i) pO2 lesser, pCO2 higher (ii) pO2 higher, pCO2 lesser (iii) pO2 higher, pCO2 higher (iv) pO2 lesser, pCO2 lesser
6. Explain the process of inspiration under normal conditions.
7. How is respiration regulated?
8. What is the effect of the partial pressure of carbon dioxide on oxygen transport?
9. What happens to the respiratory process in a man going up a hill?
10. What is the site of gaseous exchange in an insect?
11. Define the oxygen dissociation curve. Can you suggest any reason for its sigmoidal pattern?
12. Have you heard about hypoxia? Try to gather information about it, and discuss with your friends.
13. Distinguish between (a) inspiratory and expiratory reserve volume (b) inspiratory capacity and expiratory capacity (c) vital capacity and total lung capacity
14. What is tidal volume? Find out the tidal volume, to an approximate value, for a healthy human in an hour.